

**PSY504 Midterm subjective solved by S.MALIK.....**  
**11 to 23 December 2023**

**Q.1 Difference b/w similarity and good continuation?**

**Similarity**

This law says that same things are considered one thing. Similar items tend to be organized together.

**Good continuation**

The tendency to perceive a line that starts in one way as continuation in the same way.

**Q.2 Brief experiment of Hubel & Wiesel?**

Another important experiment was conducted by Hubel & Wiesel in 1962. They started their experiment with cats. They used visual cortex of the cat. They showed different stimuli, lights and shapes to cats and then they viewed cortical cells are more complex. They found that visual cortical cells responded in a more complex manner than lower cells. They named those cells bar detectors and edge detectors.

Edge detectors help us to understand where an object ends and other starts. Edge detectors respond positively to light on one side of line and negatively to light on the other side. And Bar detectors respond positively to light in the center and negatively to light at the periphery, or vice versa. Both bars and edges detectors are specific with respect to position, orientation, and width. That is, they respond only to stimulation in a small area of visual field. Bar and Edge detectors combine to see a lot of objects as shown in the figure.

**Q.3 Lobes of brain?**

The brain can be divided into four lobes:

- Occipital lobe
- Frontal lobe
- Temporal lobe
- Parietal lobe.

In each lobe are performed certain specialized functions.

Several attached pictures describe different features of the neuron, synapses, and brain organization. Study these pictures to get a broad idea of how the brain structure is organized.

## Q.4 Experiment of shriffen and Fisk?

Schneider & Fisk (1982), Schneider & Shiffrin (1977), Shiffrin & Dumais (1982), and Shiffrin & Schneider (1977) performed a series of experiments contrasting controlled vs. automatic processing.

They said automatic processes complete themselves without conscious control by the subject. In the visual and auditory report tasks reviewed earlier, the registering of the stimuli in sensory memory is an automatic process. Many aspects of driving a car and comprehending language appear to be automatic. Controlled processing seems to require conscious control. Many higher cognitive processes, such as performing mental arithmetic, are controlled. In their experiments, subjects were required to scan visual arrays. The subjects were given a target letter or number and are instructed to scan a series of visual displays for the target.

## Q.5 Subsidiary Task?

A task that typically measures how quickly people can react to target stimulus in order to evaluate the capacity demands of the primary task.

## Q.6 Time frame in the psychological time reference?

### Psychological time

We all are living in the past. This past is not in passive term. Because we receive visual information that is one second old. And we receive auditory information that is 5 seconds old. Information is not only being received but also is being reconstructed. We are creating the reality afresh, every moment. Every icon is separate entity. New information is different entity. It reconstructs reality with combining other entity. The heart of cognitive psychology lies in its experimentation.

## Q.7 Ulric Neissers experiments?

Ulric Neisser wrote first cognitive psychology book in 1957. He devised a word icon. It is a brief visual memory revealed in these experiments. He devised the word iconic memory for short term visual memory.

According to Neisser, the visual memory is neither short term memory nor long term memory but it is very very short term memory and should be called as iconic memory. Without such a visual icon, perception would be much more difficult. Many stimuli are of very duration. In order to recognize them, the system needs some means of holding on to them for a short while until they can be analyzed.

Neisser also reported that if another display is given during that one second when you are already retaining an image it is like erasing or washing out the first and overwriting the second one.

Almost all the information is held for a very brief period (1 second). It is quickly washed out after removal of stimulus unless attention is paid to it.

## Q.8 Three experiments of road accidents?

**Kahneman, Ben-Ishai & Lotan (1973) studied bus drivers.**

**Accident prone drivers:** 2 or more severe accidents in one year

**Accident-free drivers:** no accidents in the same period

**Intermediate drivers:** in-between the two groups

Selective Listening task had a high correlation with driver safety. Those drivers performed best in selective listening task they were safe drivers and had low rate of accidents.

Other road accident experiment

**Mihal & Barrett (1976)** used seven tests to predict accident involvement of commercial drivers. They found selective listening task to be the best predictor. Surprising result was because a visual task was not as good a predictor as listening. Perhaps because selective listening task is a general measure of attention.

Those good at switching attention in an auditory task are also good at visual tasks (Hunt, Pellegrino & Yee, 1989).

## Q.9 When is a skill automatic?

According to Posner and Snyder (1975)

A skill is automatic if it:

- 1) Occurs without intention
- 2) Does not give rise to conscious awareness
- 3) Does not interfere with other mental activities

## Q.10 Difference b/w Brain-injured Humans and Brains of Dead People?

The study of brain injured humans has greatly enriched our understanding of human cognition. It has allowed psychologists to design split brain experiments which made us aware of the differences between the right and the left hemispheres of the brain.

### Brains of Dead People

The study of brains of dead people has also added to understanding of cognition but to a limited extent. The brains of people with certain brain disorders were studied to see if any traces of the illness can shed light on normal brain functioning.

## Q.11 Difference b/w Synapse and neurotransmitter?

The neuron is not directly connected to other neurons. A fluid called the neurotransmitter moves between the dendrites from one neuron and the axon of the other neuron. The gap between the neuron which contains the neurotransmitter is called the synapse. It is the synapse which transmits the electric impulse generated in one neuron to the other neurons.

## Q.12 Difference b/w hardware and software level description?

The **hardware level description** may consist of studying the visual sensation itself. And we can study how the sensory neurons carry the information higher in the system. We can study the visual cortex, the part of the brain concerning itself with visual processing.

**A software level description** would start at the sensation but then continue with sensory storage, discuss the possibility of a filter underlying selective attention, then move on to short term/working memory which will process and transform the material into something that can be kept in the long term memory.

### **Q.13 What is attention?**

Attention is conceived of as being a very limited mental resource. Attention is a demanding task. It has a capacity to perform only one demanding task. Attention can't perform two demanding tasks simultaneously.

Numerous metaphors can help us to think about the limited-resource characteristics of attention.

#### **Some common metaphors;**

Is it a spotlight? A beam? (On stage)

One person is standing on stage and the beam light is just fallen on him not on whole stage.

Is it a filter? A sieve (dirt is thrown, flour)

Like you are standing in a party you just listen those people in which you are standing. Guest tunes in to one message and filters out others.

Is it a bottleneck? a narrow lane? (not enough liquid, traffic can flow).

### **Q.14 Dichotic Listening Tasks**

Cherry (1953); Moray (1959) conducted an experiment on how subjects select what sensory input they attend to. This experiment has involved a dichotic listening task.

In dichotic listening experiment subjects wear a set of headphones. They hear two messages, one ear presented with one message, the other ear with another message. Subjects pay attention to (shadow) one message and tune out the other. Psychologist discovered that very little about the unattended message is processed in a shadowing task. Subjects cannot tell what language was spoken or report any of the words spoken even if the same word was repeated over and over again. Subjects reported hearing very little from the other ear.

### **Q.15 Reaction Time & Accuracy**

The time it takes for the subject to respond to a stimulus is called reaction time. The longer the time taken, the more difficult the task. If the task is demanding, subjects likely to make mistakes and you take long time. Like mathematics is difficult subject. So we make many mistakes in mathematics because it is demanding. If there is more demanding task, the more mistakes and less accuracy. Less demanding task more accuracy. Some time reaction time slow but there is accuracy.

### **Q.16 Treisman's experiment**

Treisman (1960) conducted an experiment to know are the attention is early selector or late selector. In this experiment subjects are asked to shadow message in Left ear. The messages are:

- In Left ear: I was going there when/ China, smoke, lovely, chirping
- In Right Ear: books, chairs, tables, elephant/ I saw a bright flash

Many subjects switched ears to follow the meaningful message.

### Q.17 Kahneman's Capacity Model

In this model there are many miscellaneous determinants that impact sensory system. Something happens that trigger arousal.

Arousal means some activity starts. Arousal has many manifestations. Available capacity of attention will be allocated depending on the state of arousals. So there is allocation policy. Then there are some possible activities. Like the amount of attention is paid on a task. At the same time there is a feedback loop. We evaluate how much attention is needed for the task. Then we readjust more capacity to that task.

### Q.18 Bottleneck vs. Capacity

Both models predict that simultaneous activities are likely to interfere with each other. They attribute the interference to different causes.

**Bottleneck:** according to this model same mechanism required to perform 2 incompatible tasks. It is specific. It says same mechanism is needed to those tasks.

**Capacity:** Demands of 2 activities exceed available capacity then there is problem. It is non-specific to the task. Total demands of the task, is an important variable.

Both kinds of interference occur (specific and capacity, early selection and late selection.) Both kinds of theories are necessary.

### Q.19 Early Selection vs. Late Selection

According to this theory both early and late selection can occur. Attention is flexible. We can move our bottleneck and filter. But there must be interaction between bottleneck and capacity. We can shift our bottleneck to the low level in information processing when we pay attention to the physical properties of the task. We have a capacity to switch our bottle neck from early to late or late to early. Late selection will affect the perception of primary message because more information is selected about the secondary task.

### Q.20 Intentional vs. incidental learning

**Intentional learning** occurs when we are deliberately trying to learn; incidental learning occurs when we are not, i.e. Class teachers ask children to do but they do not do themselves. Like, parents say their children do not lie but parents lie themselves. Children learn from parent's and teacher's actions rather saying.

**Incidental learning** is as effective as intentional learning for automatic processes but is less effective for effortful learning; we know that in Urdu letter "seen" occurs more often than letter "zhe" without trying to learn this information.

## **Q.21 Most common principles of gestalt theory**

### **Proximity**

The law of proximity says that items which are close together in space or time tend to be perceived as belonging together or forming an organized group.

### **Similarity**

This law says that same things are considered one thing. Similar items tend to be organized together.

### **Good continuation**

The tendency to perceive a line that starts in one way as continuation in the same way.

### **Closure**

It refers to perceptual processes that organize the perceived world by filling in gaps in stimulation.

### **Good Form**

It is a type of closure. We fill in the gaps perceive form rather than disconnected lines.

### **Symmetry**

It says that there is a tendency to organize things to make a balanced or symmetrical figure that includes all the parts.

## **Q.22 Write Biederman three stages?**

Biederman (1987) has proposed that there are three stages in recognition of an object as a configuration of simpler components.

1. Segmentation into sub-objects
2. Classify the category of each sub-object
3. Recognition as a pattern made of sub-objects

**Remember me in your Prayers.....**

**Regards**

**S.MALIK...**